Md Affan Hussain

9110051293 | hussainaffan811@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

AI-driven Backend & Cloud Developer with 3+ years of experience architecting scalable microservices, high-performance backend systems, and data-intensive enterprise applications. Proficient in C#, .NET, SQL Server, Entity Framework Core, and AWS, he has a proven track record of optimizing complex database workloads, designing multi-tenant architectures, and building resilient asynchronous processing pipelines. Leveraging expertise in RAG, Agentic AI, MCP, and Vector Databases, he develops intelligent, production-ready solutions that enhance automation, performance, and scalability. Backed by AWS Certified Developer – Associate credentials, he delivers secure, cloud-native, enterprise-grade backend solutions.

Technical Skills

Languages & Frameworks: Python, FastAPI, C#, .NET Core, Java, SQL, JavaScript, React.js, Entity Framework Core

AI & LLM Ecosystem: RAG Architectures, Agentic Workflows, LangChain, LangGraph, MCP, Vector Databases, Amazon Bedrock, Google Gemini, Claude, Local LLMs

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, ECS, EKS, API Gateway, IAM), Git, GitHub, Docker,Portainer, CI/CD Pipelines, Azure VMs, SSH, CDN

Databases: SQL Server, PostgreSQL, ChromaDB, FAISS, DynamoDB

Testing & Tools: xUnit, NUnit, Portainer, Jira, Agile/Scrum

Backend & Architecture: RESTful APIs, Object-Oriented Programming (OOP), SOLID Principles, Design Patterns, Microservices , Low-Level Design, High-Level Design.

Experience

Software Development Engineer (SDE1) Feb 2023 – Present

Byteridge *Bengaluru, Karnataka*

- **Scalable Backend & Microservices Architecture:** Designed and maintained scalable .NET Core microservices using clean architecture and domain-driven design principles, building reliable APIs and loosely coupled services capable of supporting complex enterprise workflows and evolving business requirements.
- **Multi-Tenant Architecture:** Designed and implemented a multi-tenant architecture from scratch for a microservices-based platform using database/schema isolation in SQL Server. Established tenant-aware data access and service patterns to ensure secure data segregation while simplifying tenant onboarding and reducing operational effort.
- **High-Volume & Asynchronous Data Processing:** Engineered scalable backend processing pipelines using C# background workers, queue-based architecture, batching, bulk operations, and optimized Entity Framework Core/SQL Server queries to handle high-volume transactional workloads asynchronously. Reduced processing overhead, improved application responsiveness and throughput, and achieved approximately 35% improvement in database and API performance.
- **Database Architecture & Optimization:** Architected and optimized relational data models using SQL Server and Entity Framework Core, applying indexing, query optimization, bulk operations, batching, and efficient data-access patterns to support enterprise-scale workloads and maintain application performance as data volumes increased.
- **AI/LLM & RAG Engineering:** Engineered a production-oriented RAG-based Text-to-SQL solution using Local LLMs, enabling users to interact with relational database schemas through natural-language queries. Designed the solution to retrieve relevant schema context and generate SQL through secure RESTful APIs while keeping sensitive data within the local environment.
- **Agentic AI & Engineering Automation:** Designed Agentic AI workflows using MCP and automated development tools to streamline software engineering activities such as retrieving requirements, analyzing code, assisting with implementation, testing, and development workflow automation. Explored AI-driven approaches to improve developer productivity while maintaining enterprise security and code-quality standards.
- **Software Engineering & Technical Excellence:** Championed SOLID principles, clean code, reusable components, and maintainable architecture across backend services. Contributed to code reviews, troubleshooting, performance improvements, and technical guidance to maintain high engineering standards across the development lifecycle.

Professional Certifications

AWS Certified Developer – Associate | **Link** [09, 2026]
Amazon Web Services (AWS)

Projects

AI Sales Copilot (Team MCP) | *Python, RAG, RDBMS, Vector DB, Analytics*

- Architected an end-to-end AI sales assistant featuring a dual-layer data architecture (relational and vector databases) to automate multi-format client document ingestion (PDFs, spreadsheets) and streamline onboarding.
- Developed a suite of dynamic AI features, including a real-time technical Q&A assistant (Live AI Helper), automated pre-meeting client briefings, and an AI Estimation Engine powered by historical project data.
- Integrated predictive analytics to generate AI-driven Win Probability scores, analyzing deal metrics, historical patterns, and client signals to provide actionable insights for improving conversion rates.

Autonomous SDLC Automation Pipeline | *Claude Code, MCP, Local LLMs, Jira API, GitHub*

- Engineered a fully automated development workflow that autonomously translates Jira ticket requirements directly into committed code and GitHub pull requests.
- Leveraged **Claude Code** and the **Model Context Protocol (MCP)** to orchestrate intelligent, agentic workflows for automated code generation and version control.
- Integrated **Local LLMs** to securely process proprietary code and business logic, ensuring strict data privacy and enterprise compliance without relying on external APIs.

Achievements

- Honored with the **Performance Champion (Bytemate of the Year)** at Byteridge for exceptional overall impact and performance.
- **3-time Consistent Performer & Value Champion** awardee recognized for a strong growth mindset, innovation, and exceptional project delivery.
- Solved **500+** problems on LeetCode; secured **1st place** in the Stiegen Lane Competition for time-efficient problem-solving.

Education

Acharya Institute of Technology
Bachelor of Engineering — Information Science and Engineering | CGPA: 8.5 *Aug. 2019 – June 2023*